

**Niyam Bhushan**

Orwell or Open-Source?

Firmware is the next battleground.

Have you ever clicked a photograph from a mobile phone or a digital camera? Or listened to MP3 music in a car or home stereo? The next time you do, pause to look at the device at your fingertips, and tell yourself, "It's all about software."

Running inside the camera is the most sophisticated suite of software, working hard to conjure a perfect picture from the initial burst of pixels captured by the image-sensor. Almost all cameras provide embedded tools to touch-up and enhance your images. Think of it as a Photoshop or a GIMP inside the camera. Move over to your MP3 player in the car or the home-stereo. At the most basic level, there's MP3 decoder software at work inside the box. Not content with just that, most manufacturers have started to ship special software that analyse the MP3 stream and try to synthetically restore the original splendour of sound, lost while compressing the file into MP3. Similarly, a digital photo frame comes embedded with a JPEG decoder, and several other pieces of embedded software, based on the capabilities of the frame. In almost all cases, the embedded software is proprietary, and literally locked away in its sleek shiny box.

Consuming digital

Let's zoom out to look at the larger picture here: Consumer electronics is turning digital. Today, any digital consumer device ships with embedded software. The market for such consumer electronics (and if you add industrial devices) is many orders of magnitude bigger than the PC industry. In effect, each year, tens of billions of devices ship with embedded software that's all proprietary. Aggravatingly, ordinary appliances and machines are also beating with digital hearts. Take the cars I've owned over the last ten years. They've all had digital controllers embedded inside. Ditto for newer televisions and some washing machines. Even ordinary household switches are being transformed into digital home lighting solutions. Our lifestyle is obsessed with devices that promise us more creature-comforts, and even more creature-conveniences. Any device that brings us entertainment to dispel civilized boredom is always eagerly embraced with a fetish quite similar to a childish craze for newer toys.

Alas, the problems with proprietary software embedded so ubiquitously are age-old. Indeed, they may take on quite a sinister turn. Some manufacturers invoke the draconian

Digital Millennium Copyright Act (DMCA) if you try to hack your legally-purchased device. How would you know your MP3-player in your mobile phone does not report every unauthorised track that plays on it? Can you be sure the appliances you've bought with your hard-earned money, obey you, and are not remotely controlled and monitored by unseen masters? What about ATMs and voting machines?

'Begware'

Just wait. These fundamental problems are bound to explode rather dramatically in our world. When they do, people may finally realise the value and vision the free software community has been trying to highlight since the past 20

years. GPL version 3 partly tries to address the problem, thanks to its 'Tivoization' clause. Search the Web to familiarise yourself with this term if you don't really understand it. However, Tivoization does not solve the problem, as it still applies to free software that may get embedded.

Disappointingly, a large number

of FOSS projects that may be used as embedded software have declined to use the GPLv3 so far. Leading the pack here is Linus Torvalds himself, with his Linux kernel. Even if the entire FOSS community were to adopt GPLv3, it won't solve the problem, since it's not necessary that all manufacturers embed FOSS in their devices. What's used inside all iPods, and the newer Amazon Kindle book-readers?

What's needed is a new approach that strongly advocates firmware-openness; and maybe even includes trade and government authorities to push for complete transparency. Until that happens, consumers may just find themselves reduced to 'Begware'. Consider Apple iPhone users, paying a premium amount for a fancy phone, and then begging Apple to include features they want, or software they wish to buy, from Apple's Web-store. Just think: if you can't even be the master of any device you own, we might just be living in a new digital world of insidious slavery. 

About the author:

Inspired by the vision of Osho. Copyright February 2009: **Niyam Bhushan**, freedomyugs at gmail dotcom. First published in LinuxForYou magazine. Verbatim copying, publishing and distribution of this article is encouraged in any language and medium, so long as this copyright notice is preserved. In Hindi, 'mukt' means 'free-of-cost', and 'mukt' means 'with freedom'.